

# Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Annotator alignment with regulatory standards is largely undocumented and where documented, is inconsistent with the deployment's ground-truth norms. Annotation provenance is documented for only three datasets — DEFT-2021 (manual NER and MLC) [Q25], CLISTER (multiple annotators averaged) [Q41], DiaMED (several annotators including one medical expert, with Cohen's Kappa and Gwet's AC1) [Q45, Q46], and CAS POS (automatic via Tagex 3) [Q42]. None of these involve regulatory affairs or pharmacovigilance specialists; benchmark author affiliations are academic NLP and clinical medicine [Q8]. The elicitation (A4) confirms systematic disagreement is anticipated on borderline cases. Empirical dataset evidence corroborates: CLISTER scoring patterns are inconsistent with regulatory-equivalence norms (CLISTER-D12, CLISTER-D17, CLISTER-D18); DEFT2020 annotators score legally significant qualifier additions/deletions as near-equivalent (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); QUAERO/Mantra-GSC clinical UMLS conventions tag pregnancy contraindications by entity type rather than regulatory construct (QUAERO-D16, MANTRAGSC-D4). Convergent and external validity are both threatened. The score is 2 rather than 1 because some annotation processes are documented with multi-annotator IAA (DiaMED) and CLISTER's exact-paraphrase ceiling behavior shows annotator reliability on unambiguous cases (DEFT2020-D8).

| ID           | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Q8           | "LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Q25          | "This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER." (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q41          | "CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)                                                                                                                                                                                                                |
| Q42          | "CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)                                                                                                                                                                                                                                                        |
| Q45          | "DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4) |
| Q46          | "Table 4: Inter-annotator agreement statistics. $\kappa$ is referring to Kappa Cohen and G to Gwet's AC1." (p.4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| MANTRAGSC-D4 | Vous ne devez pas prendre Pramipexole Teva si vous allaitez. — Contraindication in patient leaflet register; condition tagged DISO not as contraindication type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DEFT2020-D8  | Troubles de l'hémostase à type de maladie de Willebrand (se traduisant par un allongement du TCA, du temps de saignement et une diminution des taux du complexe VIIIc/VWF). — Identical pair, unanimous perfect score — confirms annotator reliability on exact match                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| CLISTER-D12  | L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <b>DEFT2020-D16</b> | Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked |
| <b>QUAERO-D16</b>   | Par conséquent, Refludan ne doit pas être administré à la femme enceinte ou qui allaite. — Contraindication clause with pregnancy qualifier — UMLS tags entity type but not regulatory contraindication structure                                                    |
| <b>CLISTER-D17</b>  | Métoprolol 50 mg deux fois par jour;" / "Metformine 500 mg, 1 comprimé deux fois par jour — Different INNs scored 2.0 due to structural template; regulatory judgment would score near 0                                                                             |
| <b>DEFT2020-D17</b> | Chez les patients insuffisamment équilibrés par glimepiride arrow à la dose maximale, un traitement par l'insuline peut être associé si nécessaire. — "sous stricte surveillance médicale" absent in source; critical qualifier not flagged in score                 |
| <b>CLISTER-D18</b>  | Ésoméprazole 20 mg/jour Hépatique T1/2 = 1,0 – 1,5 h / "Périndopril 2 mg/jour NR T1/2 = 17 h — Different INNs and PK profiles scored 2.0; inconsistent with regulatory equivalence                                                                                   |
| <b>DEFT2020-D20</b> | L'absorption de boissons alcoolisées est fortement déconseillée pendant le traitement (potentialisation réciproque). — Mechanism qualifier "(potentialisation réciproque)" dropped in cible; not flagged as regulatory deviation by score                            |

## Strengths

- DiaMED documents inter-annotator agreement using both Cohen's Kappa and Gwet's AC1 across two sessions [Q45, Q46]
- CLISTER uses multiple annotators with score averaging, providing variance signal that could be analyzed for borderline cases [Q41] (DEFT2020-D7)
- CAS POS annotation was validated against manual annotations at 98% precision [Q42], establishing baseline annotation quality for that subtask
- DiaMED includes one medical expert annotator [Q45]

| ID                 | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b>Q42</b>         | "CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)                                                                                                                                                                                                                                                        |
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| <b>Q46</b>         | "Table 4: Inter-annotator agreement statistics. $\kappa$ is referring to Kappa Cohen and G to Gwet's AC1." (p.4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>DEFT2020-D7</b> | Après inspection, elles ont toutes été exclues. — High inter-annotator variance on borderline pair — illustrates scoring uncertainty                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

## Checklist responses:

**OC-1:** Ground truth labels do not reflect regulatory affairs stakeholder perspectives. No annotation guidelines reference EMA/ANSM regulatory standards in the benchmark paper. Author affiliations [Q8] are academic NLP and clinical medicine institutions. The one documented medical expert (DiaMED) annotated ICD-10 classification, not the NER or STS tasks the deployment relies on [Q45].

**OC-2:** Systematic disagreement is expected per elicitation A4. Empirical dataset evidence shows specific divergence patterns: STS scoring tolerates  $2\times$ – $60\times$  quantitative differences (CLISTER-D12, CLISTER-D13, CLISTER-D14); legally significant qualifier additions/deletions are not penalized (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); different INNs receive non-zero similarity scores due to structural template matching (CLISTER-D17, CLISTER-D18); clinical UMLS conventions tag contraindication populations by entity type rather than as regulatory contraindication structure (QUAERO-D16, MANTRAGSC-D4).

**OC-3:** Annotator demographics are NOT DOCUMENTED for the majority of corpora. The benchmark paper provides no Datasheet- or Data-Statement-level demographic breakdown. Documented information is limited to: CLISTER 'multiple annotators' [Q41], DiaMED 'several annotators including one medical expert' [Q45], DEFT-2021 'manually annotated' [Q25], CAS automatic Tagex 3 with manual validation [Q42]. Professional backgrounds beyond 'medical expert' (DiaMED) are not characterized.

**OC-4:** Re-annotation by a representative regional pool (regulatory affairs specialists, EMA/ANSM-trained pharmacologists, regulatory legal experts) would be required to align ground truth with the deployment's interpretive norms. This is consistent with elicitation A4's expectation of systematic disagreement on borderline cases.

**OC-5:** Aggregation methods: CLISTER averages annotator scores into a floating-point reference [Q41], which can erase minority/borderline disagreement. DEFT2020-D7 illustrates high inter-annotator variance (scores [2.0, 4.5, 0.0, 1.0, 0.0]) collapsed to mean 1.5, hiding the disagreement that would be diagnostic for regulatory borderline cases.

**OC-6:** Label issues harming convergent and external validity: STS labels do not converge with regulatory-equivalence judgments (CLISTER-D12–14, DEFT2020-D16–D20); NER labels do not converge with regulatory-entity granularity (QUAERO-D10–D12, MANTRAGSC-D5, MANTRAGSC-D14); aggregation by averaging hides borderline disagreements that are precisely the deployment's high-stakes flag cases.

| ID           | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| MANTRAGSC-D5 | 1 ml de solution contient 40 microgrammes de travoprost et 5 mg de timolol (sous forme de maléate de timolol) — INN and salt excipient form both tagged CHEM; no INN/excipient distinction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>QUAERO-D10</b>    | Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction                                                                                                                                                   |
| <b>CLISTER-D12</b>   | L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)                                                                                                                                                                   |
| <b>CLISTER-D13</b>   | Le taux de PSA était de 218 ng/ml (normale ≤ 4ng/ml)." / "Le dosage de PSA était de 13327 ng/ml (normal : < 4 ng/ml). — 60× magnitude difference scored 3.0                                                                                                                                  |
| <b>CLISTER-D14</b>   | Le taux de PSA était de 4,49 ng/ml." / "Le taux de PSA total était à 66,68 ng/ml. — 14× numeric discrepancy scored near-similar                                                                                                                                                              |
| <b>MANTRAGSC-D14</b> | Renagel 800 mg sevelamer — Brand name and INN both CHEM; no proprietary/INN label distinction critical for regulatory workflows                                                                                                                                                              |
| <b>DEFT2020-D16</b>  | Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked                         |
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| <b>WEB-11</b>        | Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/</a> )                                                |
| <b>WEB-12</b>        | Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap ( <a href="https://link.springer.com/article/10.1007/s40264-020-00996-3">https://link.springer.com/article/10.1007/s40264-020-00996-3</a> ) |

## Information Gaps

- No demographic breakdown of annotator pools across the majority of corpora.
- No published comparison study of biomedical NLP annotators vs. regulatory affairs specialists on the same regulatory text — confirmed field-level gap [annotator\_alignment\_verification, NOT FOUND].

## Requires Expert Verification

- Magnitude of disagreement between benchmark labels and regulatory affairs judgments on a sampled set of QUAERO EMEA / Mantra-GSC fr\_emea spans.
- Whether DiaMED's IAA methodology could serve as a template for re-annotating regulatory-relevant subsets with regulatory affairs annotators.

## Remediation

**Gap 1: Annotator demographics and professional backgrounds are undocumented for most corpora; no regulatory affairs annotators are documented anywhere in the benchmark [Q8, Q41, Q45].**

**Recommendation:** Conduct a calibration study: have a panel of regulatory affairs specialists re-score a stratified sample of CLISTER, DEFT2020 drug-leaflet pairs, and QUAERO EMEA spans against benchmark labels. Quantify systematic disagreement and use the magnitude to weight or correct benchmark performance estimates.

| ID        | Evidence                                                                                                                                                                                                                                                                                   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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